

Node Selector

Simple Definition

A Node Selector is a simple way to tell Kubernetes to run a Pod on a specific Node using labels.

Easy Memory

 Node Selector = Select a Node by its Label

Real-Life Example

Imagine a school has three classrooms.

Classroom A → Science

Classroom B → Computer

Classroom C → English

A Computer student should go only to the **Computer classroom**.

How does the student know?

Because the classroom has a **label**:

Computer

The student says:

"I want the classroom with the label 'Computer'."

This is exactly how a **Node Selector** works.

Kubernetes Example

Think of:

-  Cluster = School
-  Node = Classroom
-  Pod = Student
-  Label = Classroom Name

-  Node Selector = Student's Preference
-

How Node Selector Works

Step 1: Label the Node

Suppose you have two Nodes.

Node 1

Label:

disk=ssd

Node 2

Label:

disk=hdd

Step 2: Pod Uses Node Selector

The Pod says:

nodeSelector:

disk: ssd

Now Kubernetes checks the Nodes.

Which Node has:

disk=ssd

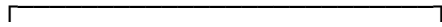
Answer:

 Node 1

So the Pod runs on Node 1.

Visual Diagram

Kubernetes Cluster



Node 1

Node 2

disk=ssd

disk=hdd



Node Selector

disk=ssd



Pod Runs Here 

Why Do We Use Node Selector?

Sometimes different Nodes have different hardware.

Example:

Node 1

SSD Storage

Node 2

GPU

Node 3

High Memory

Some applications require SSD.

Some require GPU.

Node Selector helps Kubernetes place the Pod on the correct Node.

How to Add a Label to a Node

Command

```
kubectl label nodes node1 disk=ssd
```

Meaning:

- Node Name = node1
 - Label Key = disk
 - Label Value = ssd
-

View Node Labels

```
kubectl get nodes --show-labels
```

Example Output:

NAME	LABELS
------	--------

node1	disk=ssd
-------	----------

node2 disk=hdd

Pod YAML Using Node Selector

apiVersion: v1

kind: Pod

metadata:

name: nginx

spec:

nodeSelector:

disk: ssd

containers:

- name: nginx

image: nginx

Meaning:

Run this Pod only on the Node with the label disk=ssd.

If No Matching Node Exists?

Suppose:

Node 1

disk=hdd

Node 2

disk=hdd

Pod requests:

disk=ssd

Kubernetes cannot find a matching Node.

Result:

✗ Pod remains in **Pending** state.

Node Selector Flow

Pod

|

|



Node Selector

disk=ssd

|



Scheduler

|



Find Node with

disk=ssd

|



Found?

Yes 



Run Pod

No 



Pending

Kubernetes Commands

1. View Nodes

```
kubectl get nodes
```

2. Show Labels

```
kubectl get nodes --show-labels
```

3. Add Label

```
kubectl label nodes node1 disk=ssd
```

4. Remove Label

kubectl label nodes node1 disk-

Notice the - at the end.

5. Describe Node

kubectl describe node node1

Shows all labels.

6. Create Pod

kubectl apply -f pod.yaml

7. Check Pod

kubectl get pods -o wide

Shows which Node the Pod is running on.

Node Selector vs Taints & Tolerations

Node Selector

Uses **Labels**

Applied in **Pod YAML**

Tells Kubernetes where the Pod **should run**

Simple scheduling

Taints & Tolerations

Uses **Taints** and **Tolerations**

Taint on Node, Toleration on Pod

Controls where the Pod **can or cannot run**

Advanced scheduling

Easy Comparison

Imagine a hotel.

Node Selector 

Guest says:

"I want **Room 101.**"

(Selects a room.)

Taints

Room has a sign:

VIP Only

Toleration

Guest has a VIP card.

Now the guest can enter.

Quick Revision Notes

Node Selector

- Uses **Node labels**.
- Selects a specific Node.
- Simple scheduling method.
- Pod runs only on a Node with the matching label.

Commands

```
kubectl label nodes node1 disk=ssd
```

Add a label.

```
kubectl get nodes --show-labels
```

View labels.

```
kubectl describe node node1
```

View Node details.

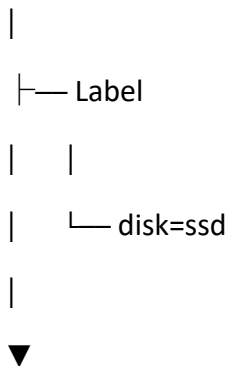
```
kubectl get pods -o wide
```

See which Node the Pod is running on.

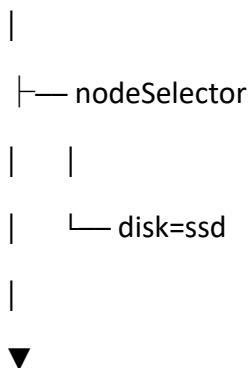
Easy Memory Trick ☆

Node Selector

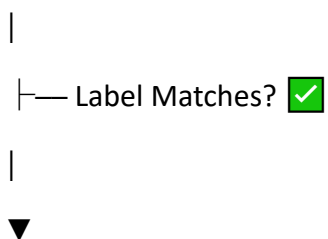
Node



Pod



Scheduler



Run Pod

One-Line Memory Trick (KCNA)

- 🏷️ Node Selector = Uses Labels to choose a specific Node.

-  Taints = Restrict Nodes.
-  Tolerations = Allow Pods onto restricted Nodes.